Supplementary Information

Laser-Assisted Nuclear Thermal Propulsion for Cislunar Logistics: A Monte Carlo Power-Closure Assessment

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S1. Thermochemistry

T (K)	p (MPa)	Reference	This model	Difference
2,800	3.0	0.0075	0.0079	5.1%
3,000	0.1	0.0793	0.0836	5.5%
3,500	3.0	0.0542	0.0577	6.5%
4,000	0.1	0.6260	0.6534	4.4%
4,500	3.0	0.3028	0.3264	7.8%
5,500	3.0	0.6982	0.7373	5.6%
6,000	3.0	0.8275	0.8601	3.9%

Table S1. Dissociation degree of hydrogen: this model and Cantera 3.2.0 (h2o2.yaml).

T (K)	p (MPa)	Gibbs minimization	Analytic
3000	0.1	0.083622	0.083622
4500	3.0	0.326411	0.326411
6000	3.0	0.860125	0.860125

Table S2. Dissociation degree from a numerical minimization of the Gibbs energy and from the analytic equilibrium constant (same thermodynamic data).

S2. Power and laser

Thrust (kN)	1300 s	1400 s	1500 s
20	217 (161–350)	267 (196–432)	316 (231–490)
50	543 (404–874)	668 (490–1,079)	790 (578–1,225)
80	868 (646–1,399)	1,069 (783–1,726)	1,263 (925–1,960)
120	1,302 (969–2,099)	1,603 (1,175–2,589)	1,895 (1,388–2,940)
150	1,628 (1,211–2,623)	2,004 (1,469–3,237)	2,369 (1,735–3,675)

Table S3. Emitted laser power (MW; median and 5th–95th percentile) for a given thrust and target specific impulse (start of the burn, full capture).

Aperture (m)	Jitter (µrad)	Range at 0.5	Range at 0.1
5	0.02	27,441	70,314
5	0.05	24,625	63,099
5	0.10	18,883	48,384
5	0.30	8,149	20,880
10	0.02	51,472	131,891
10	0.05	37,777	96,798
10	0.10	23,208	59,620
10	0.30	8,405	21,592
20	0.02	84,219	216,356
20	0.05	46,429	119,276
20	0.10	24,880	63,753
20	0.30	8,470	21,759
40	0.02	110,970	284,343

Aperture (m)	Jitter (µrad)	Range at 0.5	Range at 0.1
40	0.05	49,776	127,544
40	0.10	25,334	65,081
40	0.30	8,492	21,815
100	0.02	124,298	319,318
100	0.05	50,813	130,202
100	0.10	25,464	65,417
100	0.30	8,514	21,815

Table S4. Range (km) at which the captured fraction falls to 0.5 and 0.1 for a 3 m receiver, $M^2 = 1.2$.

Segment	Range (km)	10 m aperture	100 m aperture	Best	Aperture for 0.5 (m)	Receiver radius for 0.5, 10 m aperture (m)
LOI start	1,000	1.000	1.000	1.000	0.5	0.1
TLI end	29,000	0.691	0.880	0.882	6.3	2.3
L1 LLO	60,000	0.240	0.391	0.393	none	4.8
LEO L1	320,000	0.010	0.017	0.017	none	25.5

Table S5. Captured fraction at the ranges of the burns and relay links (0.05 µrad jitter).

S3. Mission studies

Fraction	5–95%	Median	Median Isp (s)	Closed
0.05	0.99–1.05	1.01	874	100%
0.10	0.98–1.09	1.02	891	100%
0.25	0.96–1.23	1.06	935	100%
0.50	0.95–1.47	1.13	998	100%
1.00	0.96–1.96	1.28	1,096	100%
2.00	1.01–3.20	1.62	1,256	100%

Table S6. IMLEO ratio against the useful laser power fraction (5–95%, median), median specific impulse and share of draws that close.

Receiver (kg/MW)	Design, median	Design, IQR	0.25, median	0.25, IQR
10	1.00	0.88–1.13	0.96	0.95–0.98
15	1.12	0.96–1.30	0.98	0.96–1.00
25	1.39	1.17–1.63	1.01	0.98–1.05
40	1.85	1.55–2.16	1.06	1.02–1.12
60	2.63	2.10–3.22	1.14	1.08–1.21
80	3.61	2.64–4.56	1.22	1.14–1.31

Table S7. IMLEO ratio against the receiver specific mass: median and interquartile range for the 4500 K design and for a laser fraction of 0.25.

Receiver (kg/MW)	ρ = 0.05	ρ = 0.1	ρ = 0.25	ρ = 0.5	ρ = 1	ρ = 1.5	ρ = 2	ρ = 2.4
2	0.99	0.98	0.96	0.94	0.91	0.91	0.90	0.91
4	0.99	0.98	0.97	0.96	0.95	0.96	0.96	0.97
6	0.99	0.99	0.98	0.98	0.99	1.00	1.02	1.04
8	0.99	0.99	0.99	1.00	1.03	1.05	1.08	1.12
10	1.00	1.00	1.00	1.02	1.06	1.11	1.15	1.20
15	1.00	1.01	1.03	1.07	1.15	1.23	1.32	1.40
20	1.01	1.02	1.05	1.12	1.24	1.37	1.50	1.60
30	1.02	1.04	1.11	1.22	1.43	1.64	1.86	2.03

Table S8. IMLEO ratio of the nominal vehicle on the grid of receiver specific mass and useful laser power fraction ρ.

Receiver (kg/MW)	ρ = 0.05	ρ = 0.1	ρ = 0.25	ρ = 0.5	ρ = 1	ρ = 1.5	ρ = 2	ρ = 2.4
2	385	415	538	685	907	1,198	1,499	1,831
4	459	571	766	974	1,797	3,321	4,119	5,571
6	631	838	1,050	1,936	13,581	none	none	none
8	921	1,217	2,523	32,841	none	none	none	none

Receiver (kg/MW)	ρ = 0.05	ρ = 0.1	ρ = 0.25	ρ = 0.5	ρ = 1	ρ = 1.5	ρ = 2	ρ = 2.4
10	1,073	3,792	none	none	none	none	none	none
15	none	none	none	none	none	none	none	none
20	none	none	none	none	none	none	none	none
30	none	none	none	none	none	none	none	none

Table S9. Number of missions over which the laser nodes must be amortized for the spacecraft saving to equal the node mass (none: LANTP is heavier than conventional NTP).

Time of flight (d)	Δv (km/s)	TLI burn (min)	IMLEO ratio
1.8	4.67	22	1.21
2.0	4.47	21	1.21
2.5	4.17	19	1.22
3.0	4.03	19	1.23
4.0	3.92	19	1.24
4.5	3.91	18	1.24

Table S10. Transfer time: total impulsive Δv (TLI + LOI, km/s), TLI burn time and IMLEO ratio of the nominal vehicle (receiver 10 kg/MW).

Payload (t)	NTP (t)	LANTP (t)	Ratio
5	21	38	1.86
10	31	46	1.49
20	53	64	1.21
40	101	100	1.00
80	231	174	0.75

Table S11. Payload: initial masses and ratio (receiver 10 kg/MW).

Parameter	Correlation
receiver specific mass	0.787
reactor power	0.558
chamber pressure	−0.137
nozzle recombination \phi	−0.076
tank fraction	−0.038
beam quality M^2	0.032
dropout rate	0.029
engine T/W	−0.022
re-acquisition time	0.022
pointing jitter	0.002

Table S12. Spearman rank correlation of the IMLEO ratio with the uncertain parameters.

Policy	5%	25%	50%	75%	95%	Closed
nominal (laser on both burns)	0.99	1.28	1.66	2.25	3.54	1.00
laser lost before TLI, reactor-only TLI (single fault)	1.40	1.77	2.28	3.18	4.66	1.00
reactor-only mission with the LANTP hardware (full fallback)	1.47	1.95	2.52	3.46	4.95	1.00

Table S13. Contingency policies: IMLEO ratio percentiles (5, 25, 50, 75, 95) and share of draws that close.

Fraction of burn	Propellant (t)	Burn time (min)
0.00	35.4	52
0.15	34.1	50
0.30	32.4	47
0.45	30.6	45
0.60	28.6	42
0.75	26.6	39
0.90	24.6	36
1.05	23.4	34

Fraction of burn	Propellant (t)	Burn time (min)
1.20	23.4	34

Table S14. TLI burn of the nominal vehicle after a loss of the laser at a fraction of the nominal burn time: propellant and burn time.

Payload (t)	Receiver (kg/MW)	NTP (t)	LANTP (t)	Ratio
10	10	59	76	1.28
10	25	59	140	2.36
20	10	no closure	97	–
20	25	no closure	164	–
40	10	no closure	149	–
40	25	no closure	no closure	–

Table S15. Round trip with propulsive return (four burns, reusable tug), nominal vehicle.

Output	5th	50th	95th
Isp at 4500 K design (s)	1,186.01	1,290.85	1,417.74
Effective Isp of TLI (s)	673.42	1,023.20	1,253.08
IMLEO ratio	1.02	1.69	3.70
IMLEO ratio with infrastructure	2.83	5.59	14.68
Emitted laser power (GW)	0.36	0.95	1.60
Electrical power at the node (GW)	0.89	2.34	4.31
Infrastructure mass (t)	8,281.66	19,789.63	54,805.26

Table S16. Percentiles of the main Monte Carlo outputs (1,500 draws).

S4. Studies added in versions 1.1.0 and 1.2.0

Quantity	Condition	Cantera, $\eta_n = 1$	This model, $\eta_n = 1$	Cantera × 0.94	This model, $\eta_n = 0.94$
Isp frozen	3500 K, 3 MPa	1,062	1,046	998	984
Isp frozen	4500 K, 3 MPa	1,272	1,256	1,196	1,180
Isp frozen	5500 K, 3 MPa	1,485	1,472	1,396	1,384
Isp frozen	4500 K, 7 MPa	1,255	1,235	1,180	1,161
Isp shifting	3500 K, 3 MPa	1,155	1,147	1,085	1,078
Isp shifting	4500 K, 3 MPa	1,612	1,626	1,515	1,529
Isp shifting	5500 K, 3 MPa	1,988	2,006	1,868	1,886
Isp shifting	4500 K, 7 MPa	1,509	1,516	1,418	1,425

Table S17. Specific impulse (s) of Cantera and of this model at the nozzle efficiency of 1 and of 0.94 (identical temperature, pressure and expansion ratio).

Separation (km)	Ratio	Mass (t)	TLI (min)	Max range (km)	Occulted (min)	Min Isp (s)
500	1.81	95	32	3,860	0.0	1,334
1,000	1.90	100	34	4,585	0.0	1,334
3,000	1.95	103	35	6,539	0.0	1,334
10,000	2.43	128	69	22,367	68.8	867
20,000	2.67	141	77	19,330	77.3	867
30,000	2.41	127	62	12,213	51.7	867

Table S18. Phasing of the laser node: initial separation between node and vehicle, IMLEO ratio at equal reactor power, initial mass, TLI burn time, largest node–vehicle range during TLI, time with the line of sight blocked during TLI, and lowest instantaneous specific impulse (nominal vehicle).

Burn	Mass (t)	Propellant, independent	Propellant, model	Difference	Time, independent	Time, model
TLI	40	12,869	12,893	0.18%	1,128	1,130
TLI	80	27,304	27,383	0.29%	2,393	2,400
TLI	160	61,799	61,840	0.07%	5,416	5,420
lunar departure	40	5,607	5,613	0.11%	491	492
lunar departure	80	11,382	11,410	0.24%	998	1,000

Burn	Mass (t)	Propellant, independent	Propellant, model	Difference	Time, independent	Time, model
lunar departure	160	23,782	23,823	0.17%	2,084	2,088
TLI, limiting case	40	12,534	12,550	0.13%	–	–
TLI, limiting case	80	25,067	25,101	0.13%	–	–
TLI, limiting case	160	50,134	50,156	0.04%	–	–
lunar departure, limiting case	40	5,576	5,591	0.25%	–	–
lunar departure, limiting case	80	11,153	11,159	0.05%	–	–
lunar departure, limiting case	160	22,306	22,317	0.05%	–	–

Table S19. Checks of the finite-burn integration. Rows 1–6: independent integration (adaptive DOP853) and mission-model integration (fixed step) of the same equations with constant specific impulse: propellant (kg), difference and burn time (s); this verifies the numerical accuracy only. Rows 7–12 (limiting case): near-impulsive burn of the mission model (thrust 40 times the nominal) against the rocket equation with the impulsive velocity change of the transfer (the column 'independent' then holds the rocket-equation propellant).

Parameter	First-order	First-order, 95% interval	Total	Total, 95% interval	First-order (log)	Total (log)
receiver specific mass	0.609	0.464–0.757	0.703	0.646–0.759	0.638	0.661
reactor power	0.243	0.156–0.340	0.342	0.305–0.379	0.296	0.323
chamber pressure	0.021	−0.006–0.049	0.039	0.035–0.044	0.020	0.024
nozzle recombination	0.007	−0.006–0.023	0.011	0.010–0.012	0.007	0.010
coupling efficiency	0.005	−0.009–0.019	0.011	0.010–0.012	0.005	0.007
core temperature limit	0.003	−0.011–0.016	0.007	0.006–0.008	0.003	0.004
pointing jitter	0.002	−0.005–0.010	0.002	0.002–0.003	0.001	0.002
tank fraction	0.002	−0.003–0.007	0.001	0.001–0.001	0.001	0.002
engine T/W	0.001	−0.002–0.003	0.000	0.000–0.000	0.001	0.000
nozzle efficiency	0.001	−0.001–0.003	0.000	0.000–0.000	0.000	0.000
re-acquisition time	0.001	0.000–0.002	0.000	0.000–0.000	0.000	0.000
beam quality	0.001	0.000–0.002	0.000	0.000–0.000	0.000	0.000
absorption efficiency	0.001	0.000–0.001	0.000	0.000–0.000	0.000	0.000
dropout rate	0.001	0.000–0.001	0.000	0.000–0.000	0.000	0.000

Table S20. Sobol indices of the IMLEO ratio at equal reactor power (base sample 2,048, 32,768 model evaluations, scrambled Sobol sequence): first-order index and total index, each with its 95% bootstrap interval, and the same indices for the logarithm of the ratio (last two columns).

Laser fraction	Useful power (GW)	Chamber T (K)	Receiver for ratio < 1	Receiver for ratio < 0.9
0.10	0.05	2,972	11.3	–
0.13	0.07	3,024	10.7	–
0.18	0.09	3,090	10.3	–
0.24	0.12	3,171	9.4	–
0.32	0.16	3,270	9.2	–
0.42	0.21	3,387	8.6	–
0.57	0.28	3,523	7.8	–
0.76	0.38	3,677	7.0	–
1.01	0.50	3,851	6.5	1.2
1.35	0.67	4,044	6.0	1.6
1.80	0.90	4,260	5.4	1.8
2.40	1.20	4,506	4.9	1.8

Table S21. Technology-threshold map: laser fraction, useful laser power (reactor 500 MWt), chamber temperature at that power, and the receiver specific mass (kg/MW) below which the IMLEO ratio is below 1 and below 0.9 (– : not reached down to 1 kg/MW).

Statistic	Both optimized	LANTP at the reactor of the draw
median	1.44	1.74
5th percentile	1.13	1.14
25th percentile	1.28	1.39

Statistic	Both optimized	LANTP at the reactor of the draw
75th percentile	1.69	2.37
95th percentile	1.94	3.97

Table S22. IMLEO ratio against the conventional tug with an optimized reactor (500 draws): both tugs optimized, and the laser-assisted tug at the reactor power of the draw.

Exponent x	Median ratio	5th–95th	Below 1	Optimal P, NTP	Optimal P, LANTP	M, NTP (t)	M, LANTP (t)
0.7	1.51	1.18–1.95	0%	500	100	53.2	80.1
1.0	1.49	1.16–1.94	0%	500	100	53.2	79.0
1.3	1.48	1.14–1.93	0%	500	100	53.1	78.1

Table S23. Scaling of the engine-system mass with reactor power (P_R^x at constant thrust, relative to 500 MWt) in the optimized benchmark (150 draws): median IMLEO ratio and 5th–95th percentile, share of draws below 1, median optimal reactor power (MWt) of the conventional and of the laser-assisted tug, and median initial masses (t).

Base sample	Receiver: first-order	Receiver: total	Reactor: first-order	Reactor: total
128	0.673	0.697	0.198	0.349
256	0.600	0.695	0.202	0.328
512	0.595	0.678	0.245	0.344
1024	0.600	0.702	0.248	0.345
2048	0.609	0.703	0.243	0.342

Table S24. Convergence of the Sobol indices of the receiver specific mass and of the reactor power with the base sample size.

S5. Directed-energy deflection of an asteroid (illustrative extension)

A beam of the power that LANTP would need could in principle also be used to deflect an asteroid [26]. A simple model, given as an illustration and not as a result of the transport analysis: the beam of 1.4 GW from a 10 m aperture reaches the asteroid at a range of 3×10^4–3×10^5 km, the spot size follows from the optical model of Section 4.1, the momentum coupling saturates at 30–300 μN/W with a half-saturation intensity of 10^8–10^{10} W/m², and the irradiation lasts half of a lead time of 1–10 days; the displacement achieved at the Earth is divided by the required miss of one Earth radius. The median ratio is 2×10^{-3} for a 40 m asteroid and 3×10^{-4} for a 90 m asteroid (95th percentile 3×10^{-1} and 3×10^{-2}) (Fig. S1a), and the largest diameter that is deflected is less than 1 m at 1.4 GW and 36 m at 30 GW with a 10 m aperture (Fig. S1b). The result depends on the assumed coupling and lead time and is a limitation of the concept, not a prediction.

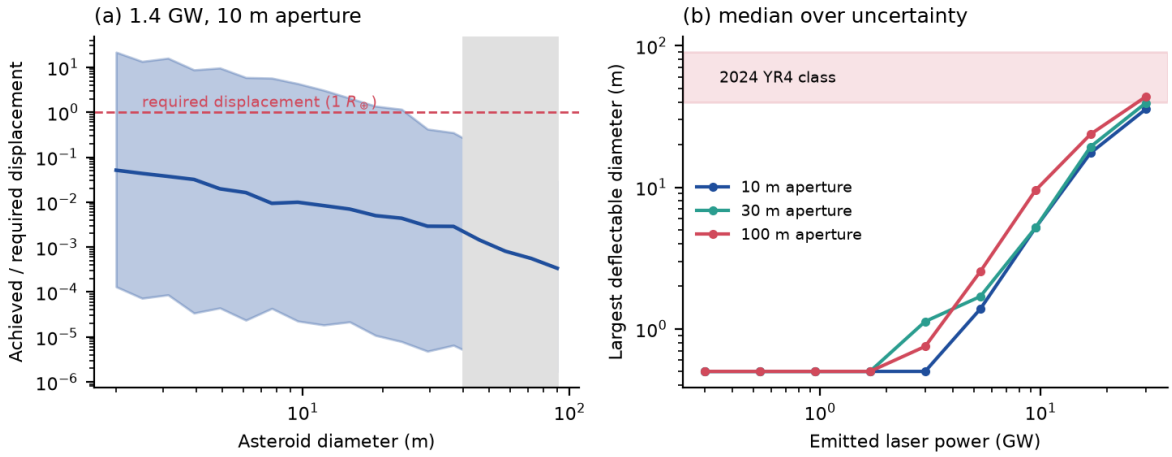


Fig. S1. (a) Achieved over required displacement against asteroid diameter for 1.4 GW and a 10 m aperture (median and 5–95% band; shaded: 40–90 m class). (b) Largest diameter that is deflected against the emitted power for three apertures.

S6. Reproducibility

The repository (https://github.com/sandlerleon/lantp-cislunar-mc, version 1.2.0) contains the model (lantp_core.py, lantp_mission.py), the studies (lantp_studies.py), the figure code (lantp_figures.py), the results (results/results.json), the automated tests (tests/test_all.py) and the manuscript sources. Commands: python lantp_studies.py --all runs every study (about 40 min on one core) and writes results/results.json and the figures; python tests/test_all.py runs the tests. Every study uses fixed seeds (base seed 2027; the Monte Carlo draws use seed 2029). Dependencies: Python 3, numpy, scipy, matplotlib; Cantera for the validation of the thermodynamics. The design inputs are the ranges of Table 2 (listed with their nominal values in results/results.json under params).